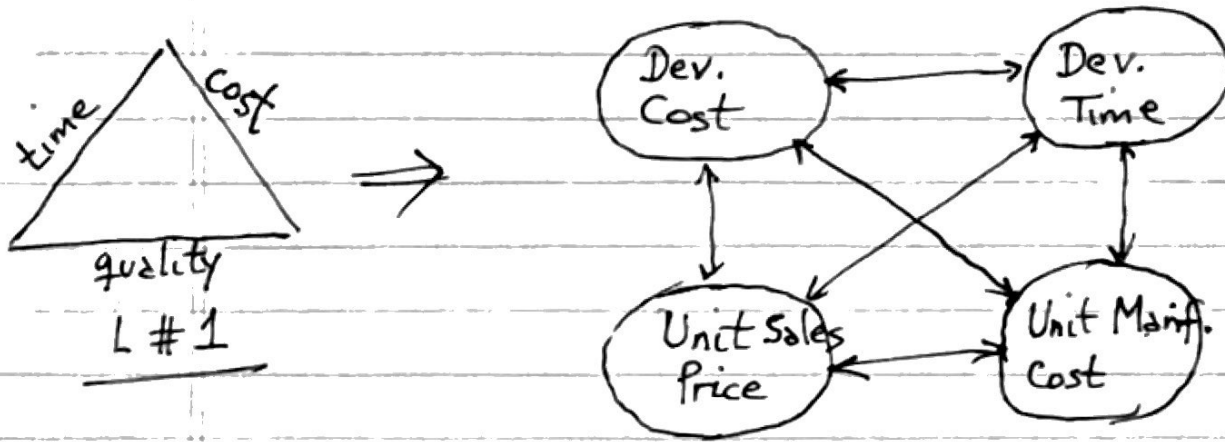


Sensitivity Analysis notes :



Effects of Trade-off on Profit

TIM 172A, Lecture # 18, 12/2/25

Agenda

1. Present Values & Net Present Value (NPV) basics
2. HW # 8
3. FMEA

1. Present Values & Net Present Value (NPV) basics

1. All future cash flows need to be converted (discounted) to present values.

2. Present value : "A dollar today (present) is worth more (i.e., has more buying power) than a dollar a year from now."

3. Therefore, all future cash flows need to be discounted when brought to the present

concept $\Rightarrow$ discounted cash flows (dcfs)
aka present values

4. The net present value (NPV) is the sum of all the DCFs or present values.

2. Calculation of present value (PV)

Consider time to be measured in periods
period : 1 (present), 2, 3, ..., n

a time period could be : 1 week, or a month
1 quarter, 1 year

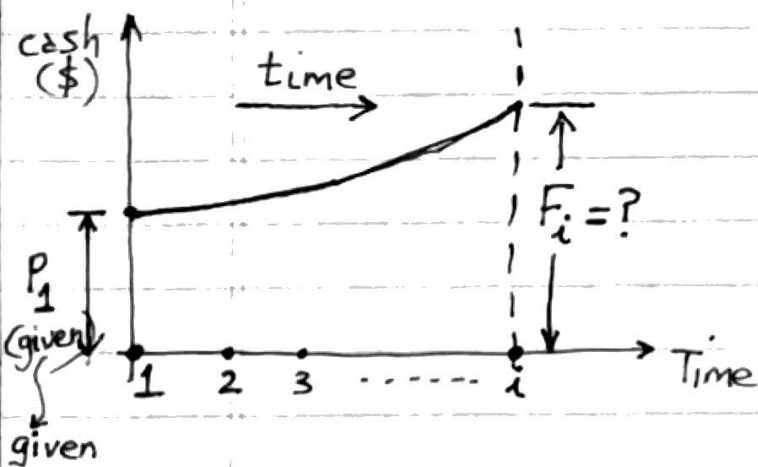
$P_1 \triangleq$ present value of cash (\$)

$F_i \triangleq$ future value of a cash flow
at period i

1 $\equiv$ present period.

Present value is simply the inverse of compound interest

Compound Interest



$$F_i = P_1 \underbrace{(1+r)(1+r)\dots(1+r)}_{(i-1) \text{ terms}}$$

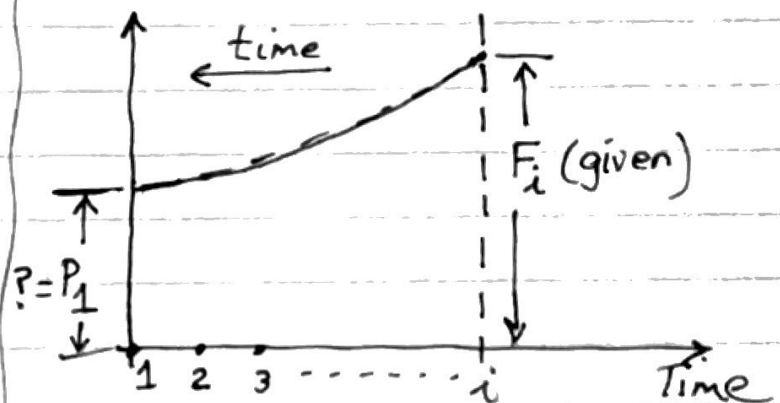
$$F_i = P_1 (1+r)^{i-1} \rightarrow (1)$$

$r \triangleq$ interest rate

$P_1 \triangleq$ principal

$F_i \triangleq$ amount of money available at a future period, i

Present Value



$$P_1 = \frac{F_i}{(1+d')^{i-1}} \rightarrow (2)$$

$d' \triangleq$ discount factor per period

$F_i \triangleq$ future cash-flow at period i

$P_1 \triangleq$ present value of F_i

$$\left. \begin{array}{l} \text{present value of} \\ \text{a future cash flow, } F_i \end{array} \right\} P_1 = \frac{F_i}{(1+d')^{i-1}}, \text{ repeated} \rightarrow (2)$$

where d' is the discount factor per period

Let one period be 1 quarter

Then $d' = \text{quarterly discount factor}$

$$\& \quad d' = \frac{d}{4}, \quad \rightarrow (3)$$

where $d \triangleq$ annual discount factor,

$$\text{and present value, } P_1 \stackrel{(2,3)}{=} \frac{F_i}{\left(1 + \frac{d}{4}\right)^{i-1}}, \rightarrow (4)$$

which is the eqn. we used for present value in L # 17 in our financial model.

3. HW # 8

Problem 1 : Intel /Apple Product Platform/Lines

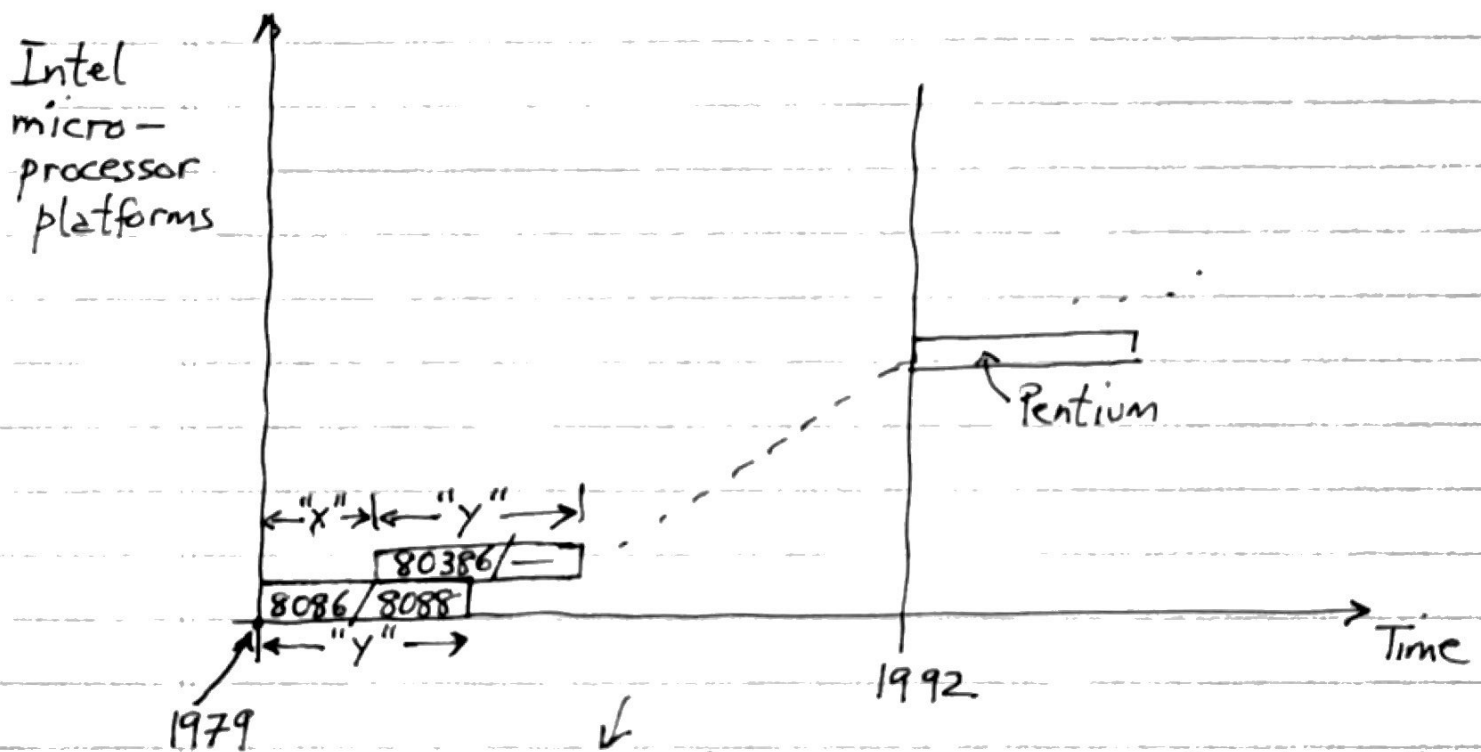
Break down the problem statements as given into a set of sub-problems; and use the SPSP to organize the solution(s).

See the "I/A ... - " handout

Intel's product platform strategy : (for the microprocessor)

Intel has a very rational product platform strategy

- Introduce a new product platform every "x" years ($x \approx 3-4$ years)
- Maintain (or support) this platform for "y" years ($y \approx 7-8$ years)



See Figure 3.8 of the "I/A..." handout

HW # 8, Prob. # 2 ("The indoor mobile robot" from HW#6)

- Use the 5-step IPMS strategy developed in L # 16
- Emulate Intel (Figures 4.1, 4.2 in the "I/A...handout")
- study the Think Jet example in L # 15